

OpenClaw RGB LED Control Tutorial

Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Switching Models], [04-AI Voice Interaction Configuration], [05-Three Chat Modes Configuration Testing]. If you are currently only using TUI / WhatsApp, you can skip `04` for now. This document assumes the basic environment is already prepared and only covers RGB-related content.

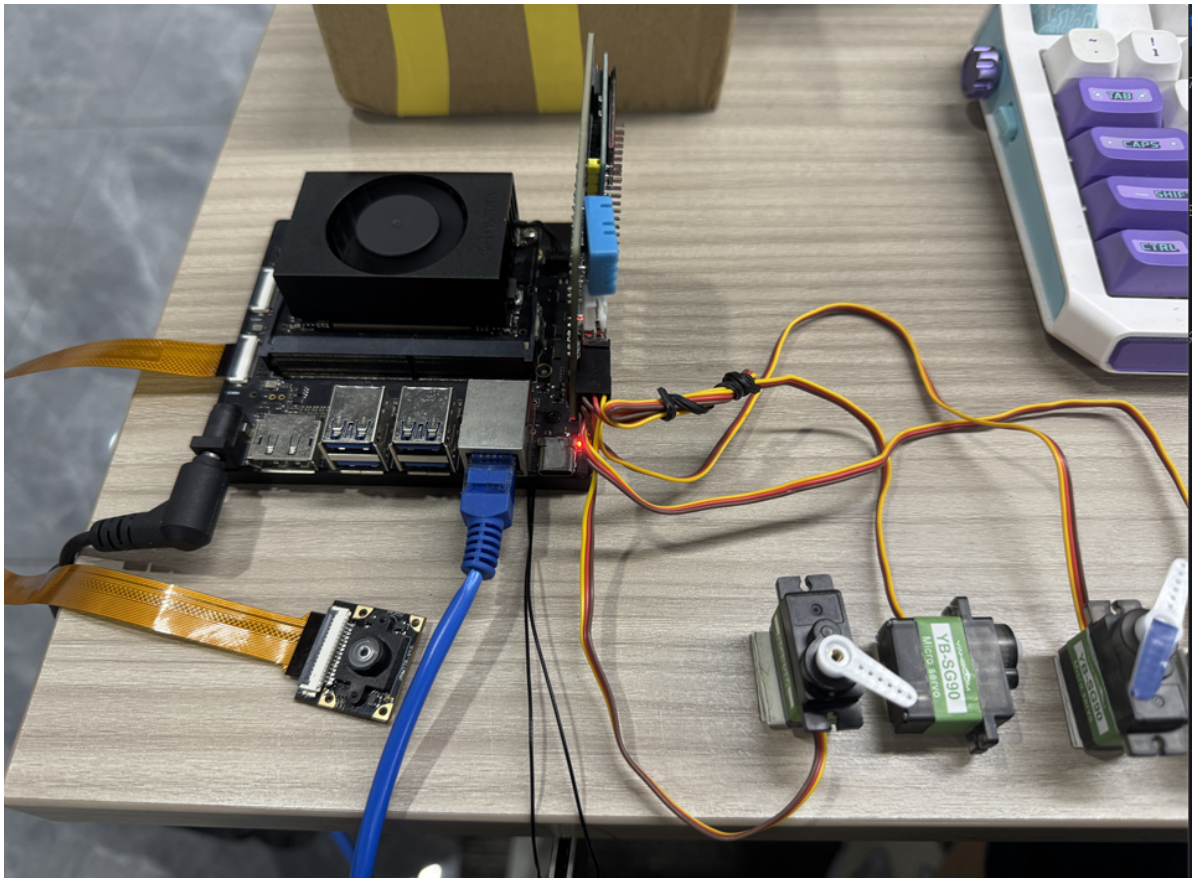
1. Tutorial Objectives

This tutorial demonstrates OpenClaw's natural language control capability through RGB LED peripherals. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw controlling RGB LEDs
- Chat with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Complete four RGB examples: single color lighting, breathing light, warning flash, rainbow marquee
- Understand the complete pipeline from "user input" to "light execution"

2. Hardware Preparation

- OpenClaw main control board
- RGB LED strip / RGB LED module [default 14 WS2812 LEDs]
- [Optional] Microphone module
- Wiring diagram



You can add your own RGB wiring diagram and hardware photo here.

3. Software Preparation

Code entry points used in this tutorial:

- Low-level driver code: `/home/jetson/hardware_hal_demo/`
- RGB control command: `/home/jetson/hardware_hal_demo/rgb`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The `rgb` tool already supports the following capabilities:

- Set fixed color `color`
- Set light effect `effect`
- Turn off lights `off`

Commonly used light effects include:

- `breathe` breathing light
- `blink` flash
- `flow` flowing light
- `marquee` marquee light
- `rainbow` rainbow light

If you want to individually verify the low-level RGB commands, you can also enter the following in the terminal:

```
rgb color --color red # All lights turn red
rgb effect --effect breathe --color blue --speed 5 # Breathing light
rgb off # Turn off lights
```

For more commands, see the markdown documentation in this folder:

```
/home/jetson/hardware_hal_demo/README.md
```

If the low-level commands can execute normally, it means the hardware connection to the Jetson Orin NX is correct and communication is established.

4. Testing RGB LEDs

Before starting this tutorial, please complete a basic `openclaw tui` chat self-check; if you haven't done so, see [05-Three Chat Modes Configuration Testing] first. After confirming OpenClaw can reply normally, proceed to the RGB-specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```

```
jetson@yahboom:~$ openclaw tui
19:04:55 [plugins] feishu_oauth_batch_auth: Registered feishu_oauth_batch_auth tool
19:04:56 [plugins] reply-bridge: Plugin registered (hooks: before_message_write, message_sending)
19:04:56 [plugins] board-control: loaded without install/load-path provenance; treat as untracked local code and pin trust via plugins.all
19:04:56 [plugins] board-control: loaded without install/load-path provenance; treat as untracked local code and pin trust via plugins.all
19:04:56 [plugins] reply-bridge: loaded without install/load-path provenance; treat as untracked local code and pin trust via plugins.all
gateway disconnected: closed | idle

session agent:main:main

what do you see

I see you sitting at a desk with a colorful mechanical keyboard – purple, teal, and white keycaps. You’re wearing a dark shirt. In the
background there are some boxes and what looks like a monitor or TV. The lighting is a bit dim.
connected | idle
agent main | session main (openclaw-tui) | baillian/mimo-v2.5 | tokens 40k/1.0m (4%)
```

You can try entering the following in the terminal:

- Set all RGB LEDs to red
- Change the RGB LEDs to blue
- Lower the RGB LED brightness a bit
- Turn off the RGB LEDs

```
Set all RGB lights to red

All RGB lights set to red. 014

Change RGB lights to blue

All RGB lights changed to blue. 015

Turn off RGB lights

RGB lights turned off. •
connected | idle
agent main | session main (openclaw-tui) | baillian/qwen3.5-plus | tokens
44k/1.0m (4%)
```

If you want to test individual LEDs, you can also try entering content like:

- Set LED 1 to green
- Set LED 5 to purple

5. Three Methods to Chat with OpenClaw

Method	Suitable Scenarios	Advantages	Recommended Positioning
WhatsApp	Remote control, classroom demo	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First get the control pipeline working
Voice Code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration method

If the AI voice module + speaker is an optional add-on, Feishu chat, TUI, and WhatsApp chat need to play back OpenClaw's replies. You can run the following program in the terminal. AI voice module chat does not need to run it again.

- Modify the broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast. After
modification, save and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) jetson@yahboom:~/voice_to_openclaw$ ^C
(.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 whatsapp Method

For WhatsApp connection steps, please refer directly to [05-Three Chat Modes Configuration Testing]. This section only retains chat examples suitable for the RGB tutorial.

After creation, you can chat directly, for example:

- Change the LED strip to blue
- Turn on the rainbow light effect
- Make the lights flash faster

19:44



< 2



+86 153 3885 7526 (你)

给自己发消息

The sweep action isn't available. I'll create the swinging motion using multiple angle changes with delays.

19:42 ✓✓

Done! The servo completed a left-to-right swing:

- Moved to 0° (left)
- Swung to 180° (right)
- Returned to 90° (center)

The servo is now ready for the next command.

19:42 ✓✓

Make the light strip blue 19:44 ✓✓

Done! The light strip is now blue at 50% brightness. All 14 LEDs are lit with a nice blue glow.

19:44 ✓✓



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123



space

return



5.2 TUI Method

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue chatting:

```
openclaw tui
```

```
OpenClaw 2026.3.12 (6472949)
Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_user_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

5.3 Voice Code Method

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake process, please refer directly to [04-AI Voice Interaction Configuration]. After waking up, you can speak your control commands directly, for example:

- Change the lights to red
- Turn on the blue breathing light
- Make the lights flow like a rainbow

```
2026-05-14 14:56:02 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 3780 ms 语音, 正在转写

[你] Take the lights red.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! The RGB light strip is now red. 🌈

All 14 LEDs are glowing red
at 50% brightness.
```

6. Comprehensive Cases

Note: The following are all demonstrated using `openc1aw tui` for chat. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: Single Color Lighting

Experimental Objective

Make the RGB LEDs consistently display a single fixed color, completing the most basic light control.

Effect Description

After receiving the command, all RGB LEDs uniformly display the specified color, such as red, green, blue, white, or warm white. This case is suitable for verifying that color mapping is working correctly.

Example Commands

- Set all RGB LEDs to red
- Change the LED strip to blue
- Change the lights to warm white

6.2 Case 2: Breathing Light

Experimental Objective

Make the RGB LEDs gradually vary in brightness on a given color, creating a breathing light effect.

Effect Description

The RGB LEDs will cycle from dim to bright, then from bright to dim. The overall effect is relatively soft, suitable for standby indicators, ambient lights, or status indicator lights.

Example Commands

- Turn on the blue breathing light
- Change the lights to a green breathing effect
- Slow down the breathing light speed a bit

6.3 Case 3: Warning Flash

Experimental Objective

Make the RGB LEDs rapidly flash to simulate a warning light or alarm effect.

Effect Description

The lights can flash rapidly using red, yellow, or alternating red and blue, suitable for expressing information like "attention", "warning", or "abnormal state", with more obvious visual feedback.

Example Commands

- Make the lights flash red
- Turn on the yellow warning light
- Make the RGB LEDs flash rapidly three times

6.4 Case 4: Rainbow Marquee

Experimental Objective

Make the RGB LEDs form a continuously flowing dynamic effect, demonstrating richer light expression.

Effect Description

The LED colors will change sequentially and flow in one direction, looking like a rainbow flowing light or marquee light. This effect is very suitable for boot animations, demonstration animations, or interactive light effects.

Example Commands

- Turn on the rainbow light effect
- Make the lights flow like a marquee
- Turn on the blue flowing light, with a faster speed